

GELO — Generalized Elo

A universal capability scale for games, reasoning, and any domain where "A beats B" (or "A solves what B can't") is measurable. GELO keeps chess Elo's constants — so a chess GELO is a chess Elo — and adds a domain-independent calibration protocol plus an intuitive "how many times better" readout.

The model

Every agent has a latent ability θ (in GELO points); every opponent/problem has a difficulty d . The probability the agent wins / solves is the logistic

$$P = 1 / (1 + 10^{-(\theta - d)/400})$$

This is simultaneously chess Elo, Bradley-Terry, and item-response theory (Rasch) — one model. What differs across domains is only *what we calibrate θ against*.

The two constants (inherited from chess Elo)

- **400 GELO = 10x better** (a "decade" of odds). $P = 91\%$.
- **120 GELO = 2x better** (one doubling / one "bit"). Since $400 \cdot \log_{10}(2) = 120.4$.

From these, the readout for any gap D :

gap D	$P(\text{win})$	odds ("x better")	out of N
120	67%	2x	wins 2 of 3
200	76%	3.2x	wins ~3 of 4
400	91%	10x	wins ~10 of 11
600	97%	32x	wins ~31 of 32
1700	~99.99%	~16000x (2^{14})	—

$\text{odds}(D) = 10^{(D/400)}$; $\text{doublings}(D) = D/120.4$.

Why the chess intuition is self-consistent (worked example)

- **"200 points = wins 3 of 4"**: $\text{odds}(200) = 10^{0.5} = 3.2 : 1 \rightarrow 76\%$. ✓

- **"top human $\approx 2^{14}$ better than a beginner"**: beginner ~ 600 , master/GM-territory $\sim 2300 \rightarrow$ gap 1700 $\rightarrow$ doublings $= 1700/120.4 = 14.1 \rightarrow 2^{14}$. ✓
- **Reasoning example**: Engine A = 1100, B = 1300 (gap 200) $\rightarrow$ B is $\sim 3.2x$ better per question: on the items that discriminate them, B is right $\sim 3x$ as often as A (76% head-to-head).

Calibration protocol ("calibrate first, then use")

1. **Reference ladder** — a monotonic set of reference opponents/difficulties we control: floor (random / novice) $\rightarrow \dots \rightarrow$ ceiling (perfect play / human-elite).
2. **Cross-table**, not single win-rates — round-robin among rungs (agents that only ever beat one opponent give a fake rating).
3. **Fit** θ for every rung by MLE (Bradley-Terry / Zermelo-MM), convert to GELO (400 = 10x).
4. **Calibration gate** — check the logistic actually fits: mean $|\text{predicted} - \text{observed}|$ pairwise win-rate. If a game's draw / first-move structure breaks the logistic, we learn that *before* quoting ratings. (Connect-4: 0.058 $\rightarrow$ fits well.)
5. **Anchor** to interpretable milestones (below), then **place agents** (next section).

Placing an agent — two equivalent routes

Step 5 can be done two ways. Both are the *same* logistic model (400 = 10x); they differ only in **what you anchor to**, and they agree by construction.

(A) Against a reference ladder — opponent-/difficulty-anchored. Play the agent against the calibrated rungs (Stockfish rungs; **ab_best** depths; MATH difficulty tiers) and read its GELO off the ladder by 1-parameter MLE. The "opponent" is a rung or a problem's difficulty. This is how the chess, Connect-4, and (via IRT over difficulty tiers) reasoning numbers here were produced. Convenient when you have a graded ladder.

(B) Pairwise, agent-vs-agent — reference-model-anchored. Two agents answer the **same** items; a judge decides each head-to-head; GELO comes from the win/loss/draw **cross-table** (Bradley-Terry), with one agent **pinned as the reference** (e.g. a strong model := 2800, or a small one := 2000) and the rest placed relative to it. Here the "players" are the *agents themselves* — the tightest analogy to chess and identical to the LMSYS-Arena rating. Convenient when you have a set of comparable agents and a meaningful reference. The **judge** is the crux: - *Verifiable domain* (math, code): the **verifier** is the judge — right beats wrong, both-right / both-wrong = draw. Objective, no model bias; **preferred whenever a checkable answer exists** (don't add a judge model where a checker suffices). - *Open-ended domain* (explanations, proofs-in-prose, "which answer is better"): a **master judge** — a model deliberately *stronger* than either contestant (e.g. a frontier model, or Kimi via an API) — decides each pair. Two caveats, both of which are just this series' thesis turned on the referee:

- (i) the rating is only as trustworthy as the judge; on items near or above the *judge's own* ceiling the **judge becomes the bottleneck** (evaluator-limited, exactly as elsewhere here);
- (ii) control judge self-preference/style bias — blind the source and randomize presentation order.

The choice of reference (2800, 2000, ...) is free — it only fixes *where the origin sits*; the constants (400 = 10×, 120 = one doubling) make every gap from it meaningful (a model 400 below the reference loses ~10:1 on the items that discriminate them). You can also **mix**: place a few agents on a graded ladder, then chain the rest pairwise off those anchors.

Interpretable anchors (the "what does the number mean" layer)

- **Chess (human population, FIDE)**: 600 novice · 1200 club · 2000 Expert · 2200 CM/NM · 2300 FM · 2400 IM · 2500 GM (~1,700 worldwide) · 2800+ world-elite.
- **Connect-4**: random = floor · **ab_best** depth-d = "d-ply tactical lookahead" · solver = perfect.
- **Reasoning**: difficulty tiers with known human solve-rates (GSM8K → MATH L1–5 → olympiad), fit by IRT; anchor θ to human-solver percentiles.

Cross-domain comparability comes from anchoring each domain to percentiles of its reference distribution: " $\theta = 2000$ " then means the same *odds vs a calibrated ladder / percentile toward the ceiling* in any domain — comparable by construction, not by analogy.

Report GELO two ways

- **GELO points** (continuity with chess: 2800 still means 2800).
- **Doublings** = GELO/120 (the " 2^N better" reading), when a bits-of-capability framing is clearer.